CentralGraphFormer: Center-Guided Spectral–Spatial Graph Transformer for Hyperspectral Image Classification

Zilin Zhang, Haoyang Yu, Zhaohui Jiang, *Member, IEEE*, Kangkang Li, Dong Pan, *Member, IEEE*, and Weihua Gui, *Member, IEEE*

***Abstract*—The hyperspectral image (HSI) classification requires the effective exploitation of spectral–spatial information under high spectral redundancy, spatial heterogeneity, and limited labeled samples. Existing CNN- and Transformer-based methods often treat surrounding pixels uniformly and rely on dense attention or shallow fusion strategies, which are poorly aligned with the inherently center-focused supervision of patch-based classification and incur high computational cost. To address these issues, we propose CentralGraphFormer, a center-guided spectral–spatial representation framework that explicitly aligns feature learning and global reasoning with the central-pixel decision objective. The framework integrates three stages: 1) a lightweight spectral shift separable network (S^3-Net) for efficient spectral continuity modeling and redundancy suppression; 2) a spatial relevance screening mechanism that adaptively selects semantically consistent neighbors via stochastic gating and biologically inspired smoothing; and 3) a central graph attention module (CGAM) that performs sparse, center-anchored global reasoning using graph-aware keys and a single center query, avoiding the dense self-attention. Extensive experiments on four benchmark HSI datasets demonstrate that CentralGraphFormer consistently outperforms state-of-the-art methods while maintaining favorable computational efficiency. Ablation studies further confirm the effectiveness and complementarity of the proposed center-guided spectral, spatial, and global modeling components.**

***Index Terms*—Graph attention networks, hyperspectral image classification, spectral shift separable convolution, spectral–spatial representation.**

I. Introduction

HYPERSPECTRAL image (HSI) simultaneously records the spatial information of surface targets and spectral reflectance across tens to hundreds of contiguous bands, forming a unique 3-D "image-spectrum" data cube. This property enables fine-grained material characterization and has supported broad applications such as precision agriculture [1], environmental monitoring [2], geological surveying [3], and defense intelligence [4]. However, the inherent characteristics of HSI data also impose key challenges for pixel-level classification. High-spectral dimensionality leads to severe redundancy and increased computational burden; spectral variability and spatial heterogeneity caused by complex surface conditions complicate consistent discrimination, and the expertise required for dense labeling results in scarce annotated data [5]. These factors collectively motivate the development of models capable of robust spectral–spatial representation learning under high dimensionality, structural complexity, and limited supervision [6].

Early hyperspectral classification research began with physics-based modeling and feature-engineering paradigms. Radiative transfer models (e.g., PROSAIL [7]) offered valuable physical interpretability but relied heavily on accurate prior knowledge and were sensitive to atmospheric conditions, restricting their practical applicability. Feature-extraction and dimensionality reduction techniques such as PCA [8], [9], and LDA [10], [11] were widely adopted to mitigate the spectral redundancy, while classical machine learning methods, including SVM [12], RF [13], and *k*NN [14], enabled more flexible decision boundaries. Although effective in controlled scenarios, these approaches typically depend on handcrafted features or predefined kernels, limiting their ability to capture the nonlinear spectral–spatial structures of complex HSI scenes.

CNN-based approaches remain a dominant paradigm in the hyperspectral image classification due to their strong inductive biases for modeling local spatial textures and short-range spectral variations [15], [16], [17]. By exploiting spatial continuity within local neighborhoods, convolutional networks effectively suppress noise and enhance discriminative feature extraction in homogeneous regions, as demonstrated by representative 3-D CNN architectures and hybrid convolutional designs [18], [19], [20]. However, their finite receptive fields and locality-driven design limit the ability to capture long-range spatial dependencies and contextual relationships across irregular or heterogeneous structures. To mitigate these limitations, recent studies have incorporated graph-based relational modeling into CNN pipelines, enabling non-Euclidean spatial–spectral relationships to be encoded alongside local convolutional features [21], [22], [23]. In this setting, graph reasoning primarily serves as a structural enhancement to convolutional

Received 30 December 2025; revised 26 February 2026; accepted 10 March 2026. Date of publication 12 March 2026; date of current version 19 March 2026. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 62573437, in part by Hunan Provincial Natural Science Foundation under Grant 2025JJ10007, and in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC1007. *(Corresponding author: Haoyang Yu.)*

The authors are with the School of Automation, Central South University, Changsha, Hunan 410083, China (e-mail: yu.haoyang@csu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3673417

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

representations through the neighborhood affinity modeling and contextual aggregation [24], [25], [26]. Nevertheless, most CNN-centric designs implicitly assume that pixels within a local neighborhood contribute with comparable importance to the classification decision, an assumption that becomes suboptimal in patch-based HSI classification, where the center pixel serves as the explicit supervision target. As a result, such approaches lack adaptive mechanisms to explicitly emphasize the center-pixel relevance under complex spatial configurations.

Transformer-based architectures further extend HSI modeling capacity by introducing global self-attention, which enables long-range interactions across spatial locations and spectral bands [27], [28], [29], [30]. By relaxing locality constraints, Transformers alleviate spectral redundancy and enhance the global contextual reasoning across spectral–spatial domains. However, dense attention mechanisms incur quadratic computational complexity, exhibit weak inductive bias toward local geometry, and often assign uniform importance to all patch pixels, resulting in sensitivity to spatial irregularities and insufficient center-pixel discrimination in patch-based classification [31], [32]. To address these limitations, recent graph-aware Transformer frameworks integrate structural priors into attention computation, enabling localized geometric relationships and topological constraints to guide global dependency modeling [33]. In these approaches, graph structures act as auxiliary guidance for attention rather than standalone reasoning modules. Despite incorporating structural priors, most existing Transformer-based solutions still rely on dense or weakly constrained interactions among patch pixels, making it difficult to explicitly align contextual aggregation with the center-pixel decision objective while maintaining computational efficiency.

Hybrid CNN–Transformer architectures seek to combine convolutional locality with attention-based global dependency modeling, offering a promising pathway toward more expressive spectral–spatial representations [34], [35], [36]. Subsequent hybrid designs further explore parallel or cascaded fusion strategies to promote sustained interaction between local convolutional features and global contextual representations [37], [38], [39], [40], [41]. Beyond purely bimodal hybrids, recent frameworks incorporate graph-based relational modeling to provide structured contextual connections and topology-aware feature aggregation [42], [43]. More recent developments extend this paradigm by integrating graph-guided objectives or multiscale structural construction, improving feature separability and robustness under limited-sample conditions [44], [45]. However, despite architectural diversity, most existing hybrid systems ultimately aggregate contextual information in a largely uniform or static manner, treating surrounding pixels as isotropic contributors and lacking an explicit mechanism for center-driven representation refinement.

These observations reveal several unresolved challenges in existing spectral–spatial learning paradigms. First, the hyperspectral image classification is inherently a center-pixel decision task; however, most existing models treat surrounding pixels uniformly, which becomes particularly problematic near class boundaries or within heterogeneous regions. Second, long-range spectral continuity and interband dependency structures are difficult to model efficiently without introducing substantial computational overhead. Third, the global feature aggregation is often conducted in an opaque manner, offering limited structural interpretability and insufficient insight into contextual relevance. Importantly, these challenges are tightly coupled: uniform neighborhood modeling weakens center-aligned discrimination, inefficient spectral modeling limits scalability, and opaque global aggregation obscures the contribution of contextual cues. Related efforts in holistic end-to-end segmentation of large-scale imagery, such as REST [46], similarly aim to capture global context efficiently via structured spatial interactions; however, unlike REST's dense spatial-parallel communication, CentralGraphFormer employs a center-guided sparse graph to selectively propagate information from semantically relevant neighbors, aligning global reasoning with the pixel-level classification objective while maintaining computational efficiency.

To address these challenges, a unified spectral–spatial representation framework, termed *CentralGraphFormer*, is presented. The proposed framework is explicitly designed around the center-pixel classification objective, adopting a progressive paradigm of "feature purification–context selection–global refinement" to align representation learning with task-specific decision requirements. Specifically, efficient spectral representation learning guided by spectral continuity priors is employed to address high-dimensional redundancy; biologically inspired spatial relevance estimation is introduced to adaptively emphasize semantically consistent neighborhoods around the center pixel; and lightweight center-driven global reasoning is realized through graph-based attention to enable interpretable and efficient contextual refinement. Through the coordinated integration of these principles, discriminative feature interactions are enhanced while computational efficiency is maintained.

The main contributions of this work are summarized as follows.

1) A spectral shift separable network (S^3-Net) is developed, in which learnable spectral shifting is embedded into a depthwise-separable convolutional structure. This design enables compact modeling of long-range and groupwise spectral dependencies while preserving the intrinsic continuity of hyperspectral signals.
2) A biologically inspired spatial relevance estimation mechanism, composed of stochastic dynamic gating (SDG) and biological vision blurring (BVB), is designed to adaptively emphasize semantically consistent neighborhoods, thereby improving robustness in heterogeneous and boundary-sensitive regions.
3) A central graph attention module (CGAM) is formulated to perform global refinement via center-driven sparse graph reasoning, providing an interpretable and computationally efficient alternative to dense self-attention for contextual aggregation.

Beyond these individual components, they are further integrated into a unified spectral–spatial learning framework that establishes a progressive pipeline for feature purifi-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

cation, context selection, and global refinement. Extensive experiments conducted on four benchmark hyperspectral datasets demonstrate the effectiveness and robustness of the proposed framework, while comprehensive ablation studies validate the complementary roles and necessity of each module.

II. Related Works

A. CNN-Based Methods

Convolutional neural networks (CNNs) represent the earliest and most mature deep-learning models for HSI classification due to their strong local inductive biases. Early designs include 2-D CNNs [16], capturing local spatial textures, and 1-D CNNs [17], processing per-pixel spectral vectors. To jointly model spatial and spectral cues, 3-D CNNs such as early 3-D CNN baselines [18] and SSRN [19] introduced volumetric convolutions, achieving substantial gains over purely spatial or spectral networks.

Subsequent developments focused on enhancing representational diversity and efficiency. HybridSN [20] combines 3-D and 2-D convolutions and has become a widely used benchmark. Multiscale and asymmetric designs such as RDACN [47] and other multiresolution CNNs [48] improve feature robustness in heterogeneous regions. Pixel-interaction networks like CNN-PPF [49] and dynamic volumetric CNNs [50] tackle small-sample challenges. Lightweight models further reduce computation via depthwise separation [51] or spectral–spatial attention gating [52]. Although these approaches enhance multilevel feature expressiveness, they typically expand spectral representations through stacked convolutions, which may preserve redundancy rather than explicitly densifying informative spectral cues.

Beyond pure convolutional architectures, graph reasoning has been incorporated to overcome CNNs' limited receptive field. CEGCN [21] employs parallel CNN and GCN branches to jointly learn local textures and long-range relations, enabling fusion of pixel-level and superpixel-level cues. Weighted fusion models such as WFCG [22] balance CNN-derived descriptors with GAT-based contextual responses, improving boundary discrimination. Serial integration strategies like FCGN [23] first extract spatial–spectral representations via CNNs and then apply graph convolution to encode structural affinities with minimal overhead.

Enhanced graph-attention variants such as CEGAT [24] stabilize training by emphasizing informative neighbors. More recent graph-aware CNN hybrids, including GAHE [25] and NAGIN [26], integrate multiscale graph attention and superpixel-guided encoding or adaptive neighborhood construction to enrich convolutional features and obtain more reliable spectral representations. These graph-enhanced CNNs demonstrate the benefit of relational priors but still rely heavily on convolution-dominant representations and weakly adaptive neighborhood aggregation. As a result, they struggle to explicitly screen spatial relevance or emphasize center-pixel importance under heterogeneous spatial configurations. This motivates models that explicitly encode center-pixel relevance while enabling the global contextual reasoning.

B. Transformer-Based Methods

Transformer architectures introduce global self-attention to model dependencies across spectral bands and spatial locations without convolution. Early HSI-specific models, such as SAT-Net [27] and SSTN [28], emphasize informative spectral bands and reduce redundancy. With ViT's success, more targeted architectures emerged: SpectralFormer [29] treats hyperspectral bands as hierarchical tokens to capture long-range spectral continuity, while Spa-Spe-TR [30] uses dual spectral/spatial branches for complementary cues. Models such as HiT [31] and LSFAT [32] incorporate learnable 3-D projections or local semantic aggregation to compensate for Transformers' weak locality.

Beyond purely attention-driven architectures, convolutional integration reinforces locality, as in MorphFormer [53] or multiscale spectral–spatial Transformers [54]. Graph–Transformer hybrids, e.g., SPGFormer [33], incorporate relational priors by jointly modeling spectral descriptors and structural topology with structure-aware attention, enabling coherent long-range spectral interactions and localized graph connectivity.

Although these graph-enhanced Transformers improve structural awareness, they still suffer from quadratic attention complexity, limited locality, and largely symmetric patch interactions. Such symmetric global reasoning paradigms are not inherently aligned with the center-pixel decision nature of hyperspectral classification, making adaptive and task-aligned discrimination difficult to achieve. This motivates hybrid architectures combining CNN locality, Transformer global reasoning, and structured center-aware attention.

C. Hybrid CNN–Transformer Methods

Hybrid CNN–Transformer architectures aim to combine convolutional locality with global self-attention. Early designs, such as CvT [34], HybridFormer [35], and SSFTT [36], inject convolutional biases into Transformer pipelines or tokenize CNN features for attention, enhancing modeling stability and spatial awareness over pure Transformers.

Parallel or cascaded dual-branch strategies followed, exemplified by HybridSN-style frameworks [37], which combine the volumetric convolution with Transformer refinement to enable sustained interaction between local spatial features and global contextual representations.

Recent hybrids emphasize cross-dimensional interaction and efficiency. CTMixer [38] blends convolution and attention mixers, DCTN [39] introduces spatial–spectral projection modules and interactive attention, GSC-ViT [40] employs groupwise separable convolution and separable attention, and ACTN [41] couples branches via adaptive relational fusion and mutual-learning supervision.

Graph reasoning has also been integrated into hybrid models. MVAHN [42] and AMGCFN [43] fuse CNN-, GCN-, and Transformer-based features to encode local textures, relational topology, and global context. More recent models, such as MCTGCL [44] and MSGTN [45] combine CNN–Transformer backbones with graph-driven contrastive learning or multiscale superpixel graphs to capture structural patterns and long-range dependencies.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

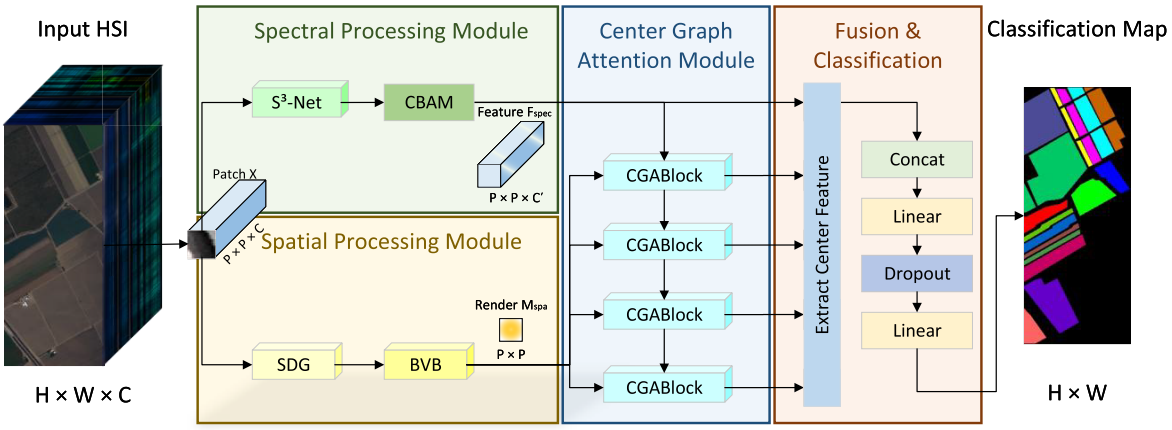


Fig. 1. Overall architecture of the proposed CentralGraphFormer. The framework consists of three major stages: spectral processing, spatial relevance screening, and center-guided graph reasoning.

Despite these advances, most hybrid systems rely on static fusion and assume uniform spatial relevance, overlooking the center-pixel decision principle. Hybrid attention modules rarely incorporate spectral continuity, and dense attention or graph operations increase computational overhead. These limitations highlight the need for hybrid frameworks that reformulate the spectral modeling, spatial interaction, and global reasoning in a center-aligned and efficiency-aware manner.

III. Methodology

A. Overview

HSI classification requires the joint exploitation of rich spectral signatures and spatial contextual information. However, simple concatenation or loosely coupled modeling of these dimensions often results in limited robustness when facing high spectral redundancy, strong interclass similarity, and spatial heterogeneity. To address these challenges, the proposed **CentralGraphFormer** adopts a center-driven, multistage architecture that explicitly aligns spectral modeling, spatial screening, and global reasoning with the supervision signal of the central pixel (see Fig. 1).

As illustrated in Fig. 1, the framework comprises three sequential processing stages. First, a *spectral processing module* performs compact spectral–spatial feature extraction using the S^3-Net, followed by a lightweight attention-based refinement. This stage targets the suppression of spectral redundancy while preserving the spectral continuity and local spatial structures critical for center-pixel classification. In parallel, a *spatial processing module* estimates the relevance of spatial positions within each patch to the central pixel via SDG and BVB, producing a smooth, center-aligned spatial relevance map.

The outputs of the spectral and spatial processing stages are subsequently fed into the CGAM, which constitutes the core global reasoning component. CGAM constructs a sparse, center-anchored relational structure and performs center-guided attention via multihead graph convolution (MHGConv), enabling long-range dependency modeling while maintaining the strict alignment with the central-pixel classification objective. Finally, center features from multiple CGAM layers are aggregated by a lightweight fusion and classification head to produce the final prediction. This overall design ensures that feature extraction, spatial relevance, and global reasoning are progressively aligned with the classification objective of the center pixel.

B. Spectral Processing Module

The spectral dimension of hyperspectral images exhibits strong interband redundancy while retaining the intrinsic continuity across adjacent wavelengths. Effectively modeling this dimension requires mechanisms that can exploit ordered spectral correlations without introducing excessive parameters or disrupting physical wavelength consistency. To this end, the proposed framework employs an S^3-Net, followed by a lightweight attention-based refinement module, as shown in Fig. 2. Although the term "shift" has been used in computer vision literature for temporal or spatial feature reindexing, the spectral shift in S^3-Net is conceptually distinct: it operates exclusively along the spectral axis to encode local interband continuity intrinsic to hyperspectral signals, rather than facilitating spatial or temporal information exchange. This design leverages the ordered and physically meaningful nature of hyperspectral wavelengths, enabling compact spectral mixing that is specifically aligned with center-pixel classification.

1) Spectral Shift Separable Network: The input hyperspectral patch is represented as $\mathbf{X} \in \mathbb{R}^{B\times P\times P\times C}$, where B denotes the batch size, $P \times P$ corresponds to a square spatial window centered at the target pixel, and C is the number of spectral channels. As illustrated in Fig. 2, the S^3-Net decomposes joint spectral–spatial modeling into three lightweight stages: depthwise spatial convolution, grouped spectral shifting, and pointwise feature fusion.

First, depthwise spatial convolution is applied independently to each spectral band to extract local spatial structures

$$\mathbf{F}_{\text{dw}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{\text{dw}}\left(\mathbf{X}\right)\right)\right) \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

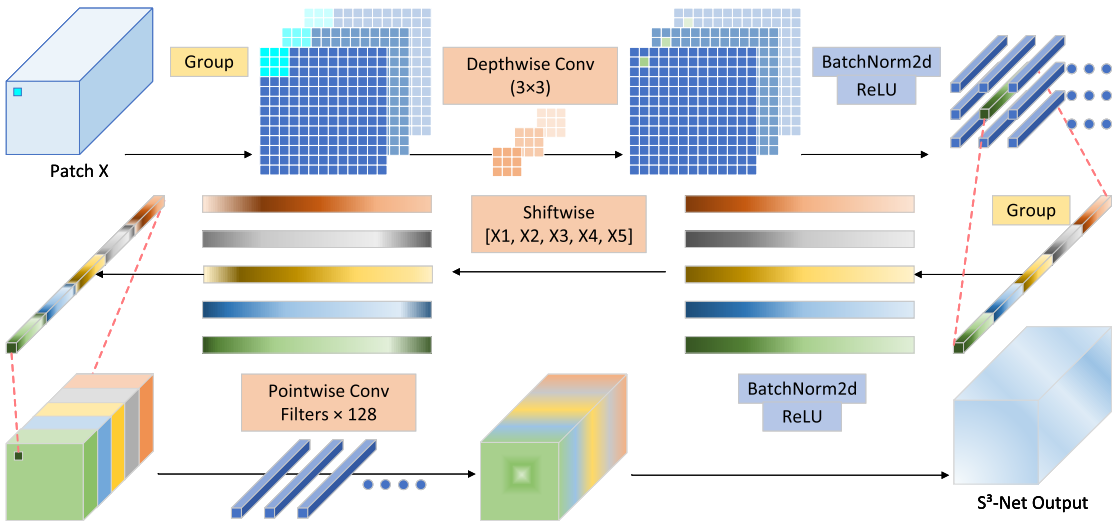


Fig. 2. Architecture of the S^3-Net. The module performs depthwise spatial convolution, grouped spectral shifting, and pointwise fusion to achieve compact spectral–spatial modeling.

where $\mathrm{Conv}_{\mathrm{dw}}(\cdot)$ denotes the depthwise convolution over the spatial dimensions. This operation preserves the original spectral dimension, ensuring compatibility with center-pixel supervision and subsequent spectral shift operations. The resulting feature tensor $\mathbf{F}_{\mathrm{dw}} \in \mathbb{R}^{B\times C\times P\times P}$ encodes bandwise spatial responses while maintaining the spectral continuity.

To explicitly model spectral continuity with minimal parameter cost, $\mathbf{F}_{\mathrm{dw}}$ is evenly partitioned into G channel groups along the spectral axis. For the gth group, a cyclic shift is performed

$$\mathbf{F}_{\mathrm{shift}}^{(g)} = \mathrm{Roll}\left(\mathbf{F}_{\mathrm{dw}}^{(g)}, \delta_g, \mathrm{dim} = 1\right) \tag{2}$$

where δ_g denotes the groupwise displacement along the spectral dimension. This grouped shift respects local spectral continuity and enables adjacent bands to interact without additional convolutional kernels, providing a physically interpretable spectral mixing strategy.

The shifted feature groups are concatenated and fused via a 1×1 pointwise convolution

$$\mathbf{F}_{\mathrm{S}^3} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}\left(\mathbf{F}_{\mathrm{shift}}\right)\right)\right) \tag{3}$$

producing a compact spectral embedding $\mathbf{F}_{\mathrm{S}^3} \in \mathbb{R}^{B\times C'\times P\times P}$, where C' denotes the reduced channel dimension after spectral fusion.

2) Lightweight CBAM Refinement: To further suppress the residual redundancy and enhance discriminative spectral–spatial cues, a lightweight convolutional block attention module (CBAM) is applied to refine the output of S^3-Net, following the sequential channel–spatial attention paradigm, as depicted in Fig. 2. CBAM provides task-adaptive spectral weighting and spatial saliency, complementing the S^3-Net's compact embedding and preparing features for center-driven spatial relevance estimation.

Given the spectral feature tensor $\mathbf{F}_{\mathrm{S}^3} \in \mathbb{R}^{B\times C'\times P\times P}$, the channelwise attention is first computed from the global average-pooled descriptor

$$\mathbf{a}_c = \sigma\left(\mathrm{Conv}_{1\mathrm{D}}\left(\mathrm{GAP}\left(\mathbf{F}_{\mathrm{S}^3}\right)\right)\right) \tag{4}$$

where $\mathbf{a}_c \in \mathbb{R}^{C'\times 1\times 1}$ assigns adaptive importance weights to different spectral channels.

Subsequently, the spatial attention is derived by applying a 3×3 convolution to the concatenation of average-pooled and max-pooled feature maps

$$\mathbf{a}_s = \sigma\left(\mathrm{Conv}_{3\times 3}\left(\mathrm{Concat}\left[\mathrm{Avg}\left(\mathbf{F}_{\mathrm{S}^3}\right), \mathrm{Max}\left(\mathbf{F}_{\mathrm{S}^3}\right)\right]\right)\right) \tag{5}$$

where $\mathbf{a}_s \in \mathbb{R}^{1\times P\times P}$ encodes the spatial saliency within the local patch.

The refined spectral representation is obtained via elementwise modulation

$$\mathbf{F}_{\mathrm{spec}} = \mathbf{a}_s \odot \left(\mathbf{a}_c \odot \mathbf{F}_{\mathrm{S}^3}\right) \tag{6}$$

where $\odot$ denotes the elementwise multiplication. The resulting tensor $\mathbf{F}_{\mathrm{spec}} \in \mathbb{R}^{B\times C'\times P\times P}$ serves as the spectral feature input to the subsequent spatial relevance screening and center-guided global refinement modules.

C. Spatial Processing Module

Patch-based hyperspectral image classification exhibits an intrinsic mismatch between the spatial extent of the input patch and the supervision signal applied only to the central pixel. Treating all spatial positions equally can introduce irrelevant or misleading context, particularly near class boundaries or in heterogeneous regions, which motivates a center-aligned relevance estimation. As illustrated in Fig. 3, the proposed spatial processing module addresses this issue by explicitly estimating spatial relevance in a center-aligned manner through two successive stages: SDG and BVB. Together, these stages produce a smooth, center-focused relevance field that selectively emphasizes informative spatial positions while suppressing background responses.

1) Stochastic Dynamic Gating: The spectral–spatial feature tensor output from the spectral processing module is represented as follows: $\mathbf{X} \in \mathbb{R}^{B\times P\times P\times C}$, where B denotes the batch size, $P \times P$ corresponds to a square patch centered at the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

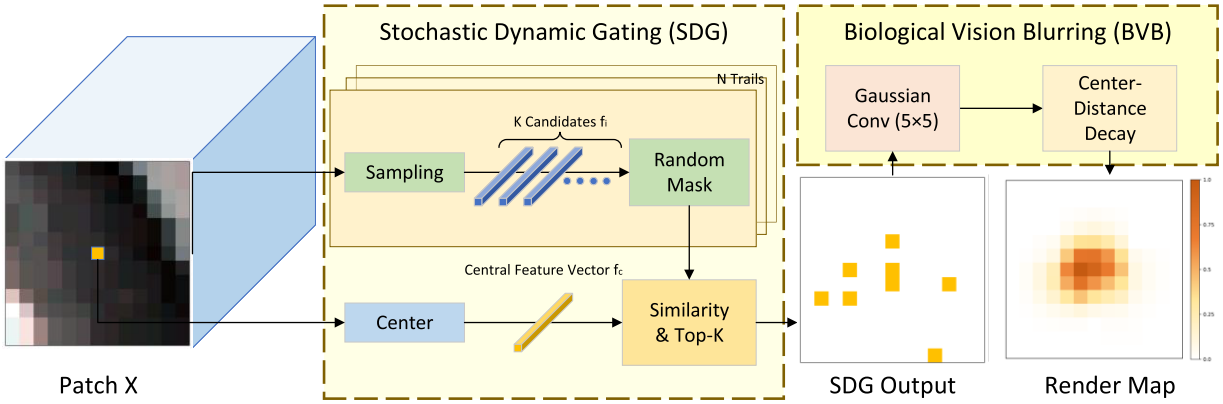


Fig. 3. Architecture of the spatial processing module. SDG selects spatial positions that are semantically consistent with the central pixel, and BVB smooths the discrete mask to produce a center-aligned spatial relevance map.

target pixel, and C is the channel dimension. As illustrated in Fig. 3, the spatial grid is then flattened into a token sequence

$$\mathbf{F} \in \mathbb{R}^{B\times N\times C}, \quad N = P^2 \tag{7}$$

where each token corresponds to one spatial position within the patch.

The central pixel is treated as the semantic reference. Its index in the flattened sequence is defined as follows:

$$i_c = \left\lfloor \frac{P^2}{2} \right\rfloor \tag{8}$$

yielding the central feature vector $\mathbf{f}_c = \mathbf{F}(:, i_c, :)$.

SDG aims to identify spatial positions that are semantically consistent with the central pixel through stochastic relevance estimation during training. The stochastic sampling introduces variability that prevents overfitting to local artifacts and enhances the generalization of spatial selection. As shown in Fig. 3, during training, in each of the T sampling rounds, two disjoint subsets of spatial positions are randomly selected as *actor* and *judge* sets. Pairwise similarity is evaluated using a fused cosine–Euclidean metric

$$S_t = \alpha S_{\cos} + (1-\alpha) S_{\text{euc}} \tag{9}$$

where $S_{\cos}$ and S_{euc} denote cosine and Euclidean similarity, respectively, and $\alpha \in [0, 1]$ balances their contributions. This combination captures both angular alignment and magnitude differences between features, providing a more robust measure of semantic consistency.

For each sampled spatial position i, its relevance to the central pixel is computed as follows:

$$r_t(i) = \alpha S_{\cos}(i, c) + (1-\alpha) S_{\text{euc}}(i, c) \tag{10}$$

where c denotes the central index. After T rounds, relevance scores are averaged to obtain $\bar{r}(i)$. The top-M positions with the highest relevance scores are retained to form a binary gating mask

$$\mathbf{M}_{\text{gate}} \in \{0, 1\}^{P\times P}. \tag{11}$$

The stochastic sampling strategy described above is applied only during training to enhance the robustness and prevent overfitting to specific spatial configurations. Random sampling is controlled by a fixed random seed to ensure reproducibility of the training process. During inference/testing, stochastic sampling is disabled. Instead, spatial relevance scores are computed deterministically for all spatial positions with respect to the central pixel, and the top-M positions are selected based on ranking without any random operation. Therefore, for a given input patch, the generated spatial mask and final prediction are fully deterministic and reproducible.

2) Biological Vision Blurring: While SDG produces a discrete and sparse relevance mask, direct use of hard gating may lead to discontinuities and sensitivity to stochastic sampling. To obtain a smooth and spatially coherent relevance field, the binary mask $\mathbf{M}_{\text{gate}}$ is further refined through a BVB process, as illustrated in Fig. 3.

First, a Gaussian smoothing kernel is applied

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{12}$$

yielding a blurred relevance map $\mathbf{M}_{\text{smooth}}$, where σ controls the spatial diffusion scale.

To explicitly encode center awareness, a radial decay field centered at $(x_c, y_c) = (\lfloor P/2 \rfloor, \lfloor P/2 \rfloor)$ is defined as follows:

$$D(x, y) = \exp\left(-\frac{(x-x_c)^2 + (y-y_c)^2}{2\sigma_d^2}\right) \tag{13}$$

where σ_d determines the decay rate with respect to spatial distance from the patch center.

The final spatial relevance map is obtained via elementwise modulation and normalization

$$\mathbf{M}_{\text{spa}} = \text{Norm}(\mathbf{M}_{\text{smooth}} \odot D) \tag{14}$$

where Norm$(\cdot)$ denotes min–max normalization. BVB smooths the discrete gating mask while emphasizing the central proximity, mitigating abrupt spatial variations and preserving biologically inspired receptive field properties. The resulting continuous relevance field

$$\mathbf{M}_{\text{spa}} \in \mathbb{R}^{P\times P} \tag{15}$$

assigns higher weights to spatial positions that are both semantically consistent with and spatially proximate to the central pixel. This relevance map is subsequently used to modulate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

patch-level features before center-guided graph-based global refinement.

D. Central Graph Attention Module

The CGAM serves as the core global reasoning component. As shown in Fig. 1, CGAM operates on spatially screened features and performs center-driven global refinement through L stacked central graph attention blocks (CGABs, whose detailed architecture is illustrated in Fig. 4). By anchoring attention to the central pixel, CGAM aligns global representation learning with pixel-level supervision, ensuring semantically relevant context aggregation.

CGAM takes as input the spectral–spatial feature tensor $\mathbf{F}_{\text{spec}} \in \mathbb{R}^{B\times P\times P\times C}$ and the spatial relevance map $\mathbf{M}_{\text{spa}} \in \mathbb{R}^{B\times P\times P}$. The spatial dimensions are flattened into a token sequence

$$\mathbf{T} \in \mathbb{R}^{B\times N\times C}, \quad \mathbf{m} \in \mathbb{R}^{B\times N}, \quad N = P^2 \tag{16}$$

where the token at the patch center serves as the query for all attention operations.

1) Center-Guided Attention With Graph-Aware Keys: The layer normalization is first applied

$$\widetilde{\mathbf{T}} = \text{LN}(\mathbf{T}) \tag{17}$$

and the central token is extracted as the query

$$\mathbf{q} = \widetilde{\mathbf{T}}(:, i_c, :), \quad i_c = \lfloor N/2 \rfloor. \tag{18}$$

For each attention head $h = 1, \ldots, H$, the MHGConv constructs a center-driven key $\mathbf{k}_h$ via

$$\mathbf{k}_h = \mathbf{D}_h^{-1/2}(\mathbf{A}_h + \mathbf{I})\mathbf{T}_h\mathbf{W}_h \tag{19}$$

where $\mathbf{T}_h$ is the head-projected center feature, $\mathbf{W}_h$ is the learnable linear mapping, $\mathbf{A}_h$ is the edge weight matrix computed by a two-layer MLP with sigmoid activation, and $\mathbf{D}_h$ is the diagonal degree matrix for normalization. The resulting key is gated via a learned sigmoid projection before participating in attention computation.

Values $\mathbf{v}$ are projected through a linear layer and optionally modulated by the spatial relevance map or rendering masks. The attention score is computed as follows:

$$\text{score} = \text{softmax}\left(\mathbf{q}\mathbf{k}^\top \Big/ \sqrt{C}\right) \tag{20}$$

and the center-enhanced token representation is obtained via

$$\mathbf{T}' = \text{score} \cdot \mathbf{v}. \tag{21}$$

2) Feed-Forward Refinement: A positionwise feed-forward network enhances nonlinearity and channelwise interactions

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2\,\text{GELU}(\mathbf{W}_1\mathbf{x}) + \mathbf{b}_2 \tag{22}$$

and the output of each CGAB is

$$\mathbf{T}_{\text{out}} = \text{LN}\left(\mathbf{T}' + \text{FFN}\left(\mathbf{T}'\right)\right). \tag{23}$$

By stacking multiple CGABs, CGAM progressively refines token representations while maintaining sparsity and computational efficiency, aligning global context with pixel-level supervision.

Fig. 5 visualizes the progressive feature refinement of CGAM on an SA dataset patch. The SDG and BVB modules guide the attention, highlighting semantically relevant spatial regions. Layerwise CGAM features show a clear evolution: early layers respond diffusely with mixed category activation, while deeper layers increasingly emphasize class-consistent patterns and suppress irrelevant background. Notably, the fourth-layer attention closely aligns with the pixel-level label distribution, demonstrating that CGAM effectively integrates the spatial guidance with center-driven global reasoning, validating the design principle of iterative context-aware feature refinement.

E. Center Feature Fusion and Classification

As illustrated in Fig. 1, the output of the stacked CGABs is further processed by a lightweight fusion and classification head. Consistent with the patch-based supervision setting, the proposed framework performs classification exclusively on the central pixel, ensuring strict alignment between representation learning and the pixel-level supervision signal. Since supervision is applied only to the central pixel, classification is performed exclusively on center features extracted from multiple refinement stages.

The token sequences produced by the L-stacked CGAB layers are denoted by $\{\mathbf{T}^{(l)}\}_{l=1}^{L}$, where $\mathbf{T}^{(l)} \in \mathbb{R}^{B\times N\times C}$ and $N = P^2$. For consistency, the spectral feature map output from the spectral processing module is reshaped into a token sequence $\mathbf{T}^{(0)} \in \mathbb{R}^{B\times N\times C}$.

For each stage, the center token is extracted as follows:

$$\mathbf{z}^{(l)} = \mathbf{T}^{(l)}(:, i_c, :) \in \mathbb{R}^{B\times C}, \quad l = 0, \ldots, L. \tag{24}$$

This formulation allows the central representation to progressively accumulate contextual information across different refinement depths, ranging from local spectral–spatial encoding to global center-guided reasoning.

Center features from all stages are concatenated and fused through a linear projection

$$\mathbf{z}_{\text{fuse}} = \mathbf{W}_f\,\text{Concat}\left[\mathbf{z}^{(0)}, \mathbf{z}^{(1)}, \ldots, \mathbf{z}^{(L)}\right] + \mathbf{b}_f. \tag{25}$$

By integrating multilevel center features, this fusion strategy enhances representation robustness and alleviates potential information loss caused by relying on a single-refinement stage.

The fused representation is then passed through a dropout layer and a fully connected classifier to produce class logits

$$\hat{\mathbf{y}} = \mathbf{W}_c\,\text{Dropout}(\mathbf{z}_{\text{fuse}}) + \mathbf{b}_c \tag{26}$$

where K denotes the number of target classes.

1) Module Summary: The proposed center-guided graph attention framework introduces a sparse yet expressive global refinement mechanism that integrates long-range spectral–spatial relationships through center-guided graph reasoning. By replacing dense attention with adaptive graph convolution and explicitly anchoring relational modeling to the central pixel, the framework avoids unnecessary interactions among irrelevant spatial tokens, CGAM achieves the efficient global context integration while maintaining the computational

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

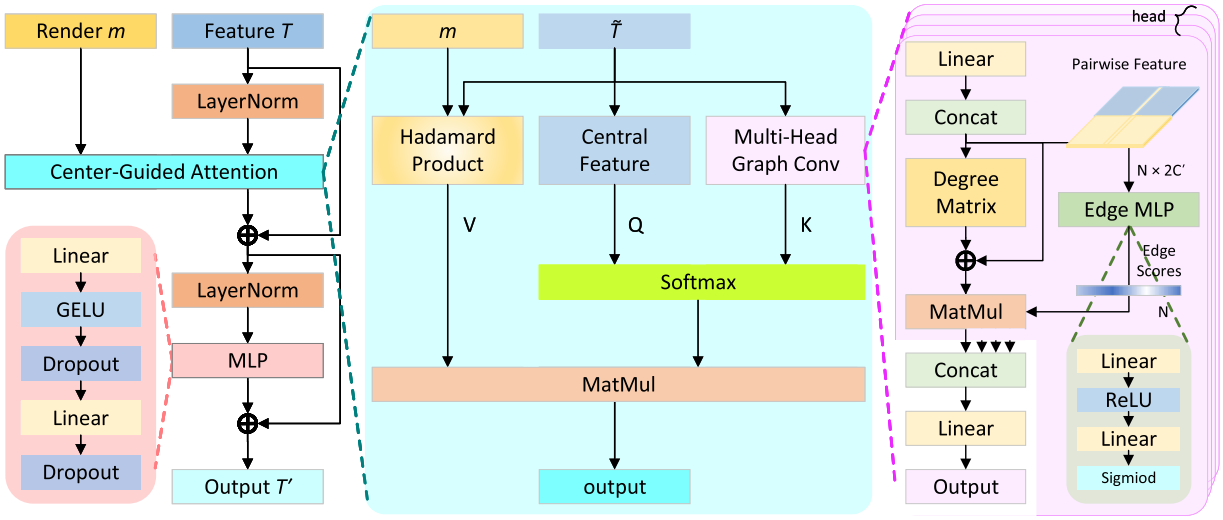


Fig. 4. CGAB structure. (Left) overall block with residual and feed-forward refinement. (Middle) center-guided attention, where a single central token computes query–key interactions with all patch tokens. (Right) MHGConv applied exclusively to the center token, producing gated keys for attention. Value features are optionally modulated by spatial relevance or rendering masks.

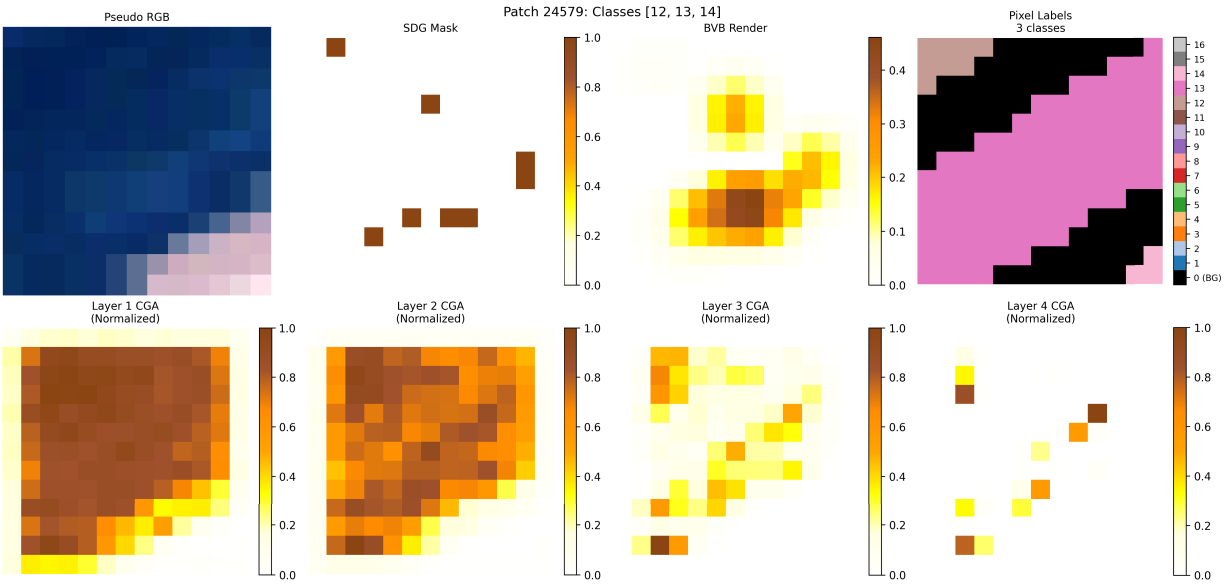


Fig. 5. Visualization of spatial module outputs and hierarchical CGAB feature maps on a patch of the SA dataset. The patch includes classes 12–14. [Top row (left to right)]: pseudo RGB image, spatial relevance (SDG) mask, BVB rendering, and pixel-level ground-truth labels, showing spatial–spectral characteristics and supervision guidance. [Bottom row (left to right)]: feature maps from CGAM layers 1–4, illustrating progressive refinement of spectral–spatial representations through center-guided attention and graph-aware key aggregation.

tractability and interpretability. Moreover, the center-driven design ensures that global reasoning directly serves the central-pixel classification objective, rather than indiscriminately modeling all spatial positions. This design enables robust feature refinement that directly supports pixel-level hyperspectral classification under limited-sample and heterogeneous conditions.

IV. Experiments

This section first describes the datasets and experimental setup. It then details the hyperparameter settings, presents an ablation study, and benchmarks the proposed method against state-of-the-art alternatives. Finally, the impact of sample-width selection and the proportion of training data on classification accuracy is analyzed.

A. Datasets Description

Four widely used hyperspectral datasets were employed in all experiments, including Kennedy Space Center (KSC), Pavia University (PU), Salinas (SA), and HanChuan (HC). For each dataset, the same training/validation/testing splits were applied consistently across all comparison experiments, ablation studies, and robustness analyses in Sections IV-C–IV-E.

1) KSC: Acquired by the AVIRIS sensor over the Kennedy Space Center, Florida, in 1996. The scene contains 512×614 pixels with 18-m spatial resolution. After removing 48 low-SNR bands, 176 bands in the range 400–2500 nm are retained. The 13 land-cover classes produce 5211 annotated samples.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NUMBERS OF SAMPLES OF THE KSC DATASET

Class ID	Class name	Train	Validation	Test
1	Scrub	71	31	519
2	Willow swamp	33	12	198
3	CP hammock	34	13	209
4	CP/oak	34	13	205
5	Slash pine	25	8	128
6	Oak/broadleaf	31	12	186
7	Hardwood swamp	19	6	80
8	Graminoid marsh	49	21	339
9	Spartina marsh	61	26	433
10	Cattail marsh	49	21	334
11	Salt marsh	50	22	347
12	Mud flats	59	25	419
13	Water	101	47	779
	Total	616	257	4176

TABLE II
NUMBERS OF SAMPLES OF THE PU DATASET

Class ID	Class name	Train	Validation	Test
1	Asphalt	126	314	5830
2	Meadows	331	824	15311
3	Gravel	39	96	1774
4	Trees	60	148	2749
5	Painted metal sheets	27	69	1249
6	Bare Soil	101	252	4676
7	Bitumen	27	68	1235
8	Self-Blocking Bricks	74	185	3423
9	Shadows	19	49	879
	Total	804	2005	37126

TABLE III
NUMBERS OF SAMPLES OF THE SA DATASET

Class ID	Class name	Train	Validation	Test
1	Brocoli green weeds 1	37	92	1695
2	Brocoli green weeds 2	75	187	3464
3	Fallow	40	99	1825
4	Fallow rough plow	28	68	1260
5	Fallow smooth	53	133	2455
6	Stubble	77	191	3539
7	Celery	70	174	3220
8	Grapes untrained	220	548	10187
9	Soil vinyard develop	123	308	5712
10	Corn senesced green weeds	63	156	2886
11	Lettuce romaine 4wk	21	51	934
12	Lettuce romaine 5wk	37	92	1688
13	Lettuce romaine 6wk	18	44	809
14	Lettuce romaine 7wk	21	51	943
15	Vinyard untrained	133	334	6177
16	Vinyard vertical trellis	26	66	1202
	Total	1042	2594	47996

2) PU: Captured by the ROSIS sensor over the University of Pavia, Italy, in 2001. The image consists of 610×340 pixels at 1.3-m resolution, with 103 spectral bands covering 430–860 nm. It includes nine urban surface classes with 42 776 labeled pixels.

3) SA: Another AVIRIS scene collected over Salinas Valley, California, in 1998. The hyperspectral cube is 512×217 pixels at 3.7-m resolution. A total of 204 bands (400–2500 nm) remain after removing water absorption channels. The 16 agricultural classes result in 54 129 labeled samples.

4) HC: A UAV-borne dataset captured by the Headwall Nano-Hyperspec imager over HanChuan, Hubei, China, in 2016. It contains 1217×303 pixels with 0.109 m resolution and 274 spectral bands spanning 400–1000 nm. The 22 fine-grained crop and man-made classes yield 2 47 572 labeled samples.

TABLE IV
NUMBERS OF SAMPLES OF THE HC DATASET

Class ID	Class name	Train	Validation	Test
1	Strawberry	200	255	42062
2	Cowpea	139	110	21593
3	Soybean	78	50	9686
4	Sorghum	55	26	5039
5	Water spinach	35	7	1158
6	Watermelon	51	22	4195
7	Greens	59	30	5808
8	Trees	117	89	17419
9	Grass	77	48	9319
10	Red roof	76	48	9257
11	Gray roof	110	81	15899
12	Plastic	47	18	3444
13	Bare soil	73	45	8625
14	Road	114	86	16764
15	Bright object	35	6	1076
16	Water	200	546	70862
	Total	1466	1467	242206

The precise per-class statistics used throughout Sections IV-C and IV-D are listed in Tables I–IV. All methods, including baselines and ablation variants, strictly follow these identical data partitions to ensure fair comparison and reproducibility.

B. Experiment Setting

All experiments reported in Sections IV-C–IV-E were conducted under a unified training and evaluation pipeline. The proposed framework was implemented in Python 3.9.23 using PyTorch 2.7.1+cu118 and executed on a workstation equipped with an Intel Xeon Gold 5318Y CPU, 256-GB RAM, and an NVIDIA RTX 3090 GPU with 24-GB memory. All baseline models were reimplemented or executed under the same software and hardware environment to ensure fair and consistent comparisons in terms of performance, runtime, and computational complexity.

1) Training Procedure: All models—including convolutional, Transformer-based, and hybrid architectures with or without graph reasoning—were optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. The batch size was fixed to 32 for all experiments. A ReduceLROnPlateau scheduler was applied with a decay factor of 0.1 to adapt the learning rate during training. Training was conducted for a maximum of 200 epochs, with early stopping enabled using a patience of 20 epochs to prevent overfitting. Cross-entropy loss was adopted as the optimization objective, and mixed-precision training with gradient scaling was employed to accelerate convergence and reduce the memory consumption.

2) Data Preprocessing: Each spectral band of the input hyperspectral images was normalized to zero mean and unit variance. During training, spatial patches of size 13×13 were extracted around labeled pixels, while supervision was applied only to the central pixel of each patch, consistent with the center-guided classification setting adopted in all comparison and ablation experiments.

3) Evaluation Protocol: To ensure statistical robustness, each experimental configuration—including baseline models, ablation variants, and the proposed method—was repeated 12 times with different random seeds. The final results are reported as the mean and standard deviation on the test set,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

using overall accuracy (OA), average accuracy (AA), and Cohen's Kappa coefficient as evaluation metrics.

4) Computational Complexity Assessment: For a fair and consistent assessment of computational cost, the number of parameters, floating-point operations (FLOPs), and GPU inference latency were measured for all models using 256 nonoverlapping 13×13 patches, ensuring that dense attention mechanisms and sparse graph-based operations were evaluated under identical preprocessing, hardware, and runtime conditions.

C. Comparison Experiments

This study evaluates the performance of the proposed method through comprehensive comparisons with four categories of baseline models, including convolutional networks (2DCNN [16], 3DCNN [18]), Transformer-based models (SSFTT [36], morphForme [53], and spectralFormer [29]), CNN–Transformer hybrids (DCTN [39], GSC-ViT [40], and MCMTN [55]), and graph convolutional networks (MiniGCN [56] and LG-GCN [57]). The experiments are performed on four widely used hyperspectral datasets (KSC, PU, SA, and HC), each averaged over 12 independent runs. OA, AA, Kappa coefficient, and per-class accuracy are reported as mean $\pm$ standard deviation to evaluate generalization, robustness, and class-level behaviors under varying scene characteristics. The per-class results provide insight into model behavior in heterogeneous spectral–spatial environments and support fine-grained interpretation from individual classes to global metrics. In addition to accuracy-based evaluation, model efficiency is assessed in terms of parameter count, computational complexity (FLOPs), memory usage, and inference time.

1) Quantitative Analysis: On the KSC dataset (the experimental results are summarized in Table V), the per-class results highlight the method's behavior in spectrally complex wetland vegetation. The most notable improvements occur in classes with pronounced intraclass variation, such as Class 4 (CP/oak), where an accuracy of $97.52\% \pm 5.19\%$ is achieved, indicating clear gains. Although MCMTN attains near-perfect accuracy in several classes, the proposed method maintains consistently high performance across all categories. Stable performance is also observed in homogeneous regions such as Class 13 (water), where $100\% \pm 0\%$ accuracy is obtained. While Class 2 (Willow swamp) exhibits slightly larger deviations, its accuracy remains high. Overall, the per-class deviations remain within moderate ranges, indicating stable predictions across diverse wetland categories.

For the PU dataset (the experimental results are summarized in Table VI), which contains materials with high spectral similarity, the per-class results show clear improvements in categories prone to confusion in urban scenes. Class 3 (gravel) reaches $97.92\% \pm 1.53\%$, demonstrating a clear advantage, while Class 9 (shadows), typically challenging due to low reflectance, attains $96.13\% \pm 2.10\%$. Across most classes, the standard deviations remain small, indicating consistent predictions under strong spectral mixing. One exception is Class 7 (bitumen), where the limited number of training samples may influence the observed deviation, suggesting that further validation with richer sampling is desirable.

In the SA dataset (the experimental results are summarized in Table VII), which exhibits strong geometric heterogeneity among agricultural categories, the proposed method achieves high accuracy across most classes, particularly those with repetitive spatial structures such as vineyards and crop rows. For example, Class 8 (grapes untrained) reaches $100\% \pm 0\%$, and Class 11 (Lettuce romaine 4wk) attains $99.19\% \pm 0.81\%$, with reduced variation compared with other methods. Overall, most classes exceed 99% accuracy, and the deviations remain low (approximately $\pm 0.35\%$ on average), indicating stable performance under varying growth patterns and field structures.

For the HC dataset (the experimental results are summarized in Table VIII) with only 0.6% training samples, the per-class results illustrate the model's behavior under extremely limited supervision. Notable improvements are observed in categories with scarce samples or high intraclass variability. Class 6 (water spinach) achieves $88.03\% \pm 2.77\%$, showing clear gains over convolutional and graph-based baselines, while Class 16 (Water) maintains a high accuracy of $99.65\% \pm 0.19\%$ with limited fluctuation. Larger deviations appear in Class 15 (bright object), likely due to sensitivity to specular reflection, but most classes exhibit controlled variability, indicating reliable performance in small-sample regimes.

In addition to classification accuracy, we report model complexity and efficiency metrics, including parameter count, FLOPs, memory usage, and inference time, to provide a comprehensive tradeoff evaluation. Compared with transformer-based baselines such as DCTN and spectralFormer, the proposed method maintains substantially lower parameter count and memory consumption while achieving higher classification accuracy. Although the inference latency is higher than that of lightweight GCN-based models, this increase is mainly due to the center-guided graph refinement and hierarchical attention operations introduced in CGAM. Importantly, the inference time varies only slightly across datasets and remains stable despite differences in scene scale and class distribution. Considering the consistent improvements in OA, AA, and κ, these results demonstrate a balanced tradeoff between representational capacity and computational efficiency.

Overall, the per-class analyses across the four datasets illustrate the method's ability to maintain stable predictions under diverse conditions, ranging from spectrally complex vegetation (KSC) and heterogeneous urban surfaces (PU) to geometrically irregular agricultural fields (SA) and extremely small-sample settings (HC). These observations complement the aggregated metrics reported in the tables and offer class-level evidence for the method's robustness across multiple remote sensing scenarios.

2) Qualitative Analysis: Qualitative classification maps on the four experimental datasets (KSC, PU, SA, and HC, whose classification maps are shown in Figs. 6–9, respectively) are used to assess spatial consistency, boundary preservation, and robustness to noise. Across datasets, the proposed method generally yields smoother region structures and clearer transitions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE KSC DATASET WITH 10% TRAINING SAMPLES

Class	2DCNN	3DCNN	SSFTT	morphFormer	spectralFormer	DCTN	GSC-ViT	MCMTN	MiniGCN	LG-GCN	Ours
1	99.58±0.18	99.76±0.25	**99.98±0.05**	99.77±0.20	99.73±0.16	99.94±0.09	99.64±0.11	100±0	99.82±0.39	99.79±0.19	99.86±0.50
2	96.29±3.84	97.94±2.70	**99.16±1.26**	97.60±3.09	97.73±2.01	97.01±4.18	95.62±3.44	96.97±4.64	65.11±35.69	98.82±1.54	98.32±2.39
3	97.13±2.62	98.17±1.83	99.72±0.56	97.85±2.62	94.22±4.97	**100±0**	98.84±1.52	99.92±0.28	40.19±44.67	**100±0**	99.76±0.38
4	88.98±2.33	84.63±7.68	91.14±6.46	92.68±5.50	63.82±13.76	95.25±2.60	80.77±8.83	93.98±4.95	60.85±33.37	88.41±6.44	**97.52±5.19**
5	71.03±7.04	75.06±5.06	75.78±9.25	61.39±10.66	38.15±20.85	81.90±1.28	71.16±7.25	**83.99±3.98**	71.75±35.55	70.64±10.08	81.51±1.92
6	86.60±4.87	93.01±2.77	99.01±1.07	93.82±4.56	89.47±6.53	**99.82±0.27**	94.27±6.73	99.73±0.63	71.01±38.17	98.07±2.03	98.97±1.66
7	99.79±0.72	99.17±1.54	**100±0**	98.65±2.89	71.77±17.35	99.06±2.33	**100±0**	99.58±1.44	65.94±45.80	96.98±3.08	99.27±1.46
8	98.48±0.66	98.92±0.93	99.63±1.28	99.24±1.54	99.68±0.34	**100±0**	95.97±2.78	**100±0**	93.07±9.22	99.98±0.08	**100±0**
9	99.39±0.20	98.62±1.47	99.94±0.10	98.67±2.16	99.90±0.33	**100±0**	98.84±1.08	**100±0**	86.47±14.24	**100±0**	**100±0**
10	98.40±0.73	97.58±1.36	99.88±0.43	98.93±1.09	99.73±0.62	**100±0**	95.91±1.97	**100±0**	92.99±12.46	99.95±0.12	**100±0**
11	97.81±0.77	99.09±0.69	99.31±0.70	99.66±0.56	99.42±1.00	**99.98±0.08**	97.10±1.44	99.26±0.57	99.98±0.08	99.45±0.68	99.83±0.29
12	**100±0**	98.81±1.35	99.44±0.94	99.76±0.34	99.16±0.93	**100±0**	97.83±2.14	99.92±0.19	98.95±2.02	**100±0**	**100±0**
13	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	99.96±0.15	**100±0**	99.84±0.06	**100±0**	**100±0**
OA(%)	97.10±0.37	97.31±0.60	98.56±0.56	97.57±0.66	94.70±1.63	99.03±0.24	96.37±0.74	98.98±0.43	87.94±4.35	98.26±0.28	**99.13±0.31**
AA(%)	94.89±0.60	95.44±0.91	97.15±1.11	95.23±1.16	88.68±3.68	97.92±0.42	94.30±1.07	97.95±0.75	80.46±7.73	96.32±0.64	**98.08±0.48**
κ (%)	96.77±0.41	97.01±0.67	98.40±0.63	97.29±0.74	94.09±1.82	98.93±0.27	95.96±0.82	98.86±0.47	86.57±4.84	98.06±0.31	**99.03±0.34**
Params(k)	1653.3	878.3	875.0	133.6	819.9	4411.5	619.3	206.7	3842.9	76.4	566.0
FLOPs(M)	1912.0	6041.5	2257.5	751.2	4845.9	7516.0	2562.1	978.2	123.0	154.0	1983.5
Memory(MB)	27.0	125.8	89.1	93.9	164.0	150.3	175.4	57.6	92.3	169.3	41.4
Inference(ms)	0.30	3.31	3.20	2.99	5.08	3.24	4.34	1.79	0.99	8.88	8.85

TABLE VI

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE PU DATASET WITH 2% TRAINING SAMPLES

Class	2DCNN	3DCNN	SSFTT	morphFormer	spectralFormer	DCTN	GSC-ViT	MCMTN	MiniGCN	LG-GCN	Ours
1	94.00±0.78	95.55±1.96	96.86±0.88	94.61±1.70	91.21±1.72	96.69±0.83	92.23±1.67	98.79±0.61	87.74±24.89	88.48±2.54	**99.65±0.55**
2	97.86±0.23	98.01±0.81	99.70±0.17	98.95±0.38	99.44±0.33	99.51±0.16	98.22±0.46	**99.86±0.08**	99.58±0.60	99.65±0.38	99.67±0.59
3	74.27±2.01	58.35±6.10	82.21±6.83	83.01±4.97	76.62±3.54	89.60±3.99	74.33±3.50	94.52±1.46	64.97±36.14	76.40±5.91	**97.92±1.53**
4	93.47±0.39	96.35±0.88	97.98±0.45	94.18±1.73	94.05±1.77	96.61±0.67	93.97±1.48	98.01±0.55	**99.88±0.12**	89.32±2.03	99.73±0.32
5	99.69±0.36	94.87±2.41	99.98±0.07	99.97±0.07	99.94±0.14	**100±0**	96.01±1.79	**100±0**	95.60±15.08	99.87±0.27	99.96±0.07
6	90.98±1.08	87.98±3.95	**99.43±1.01**	98.79±1.05	95.91±1.36	99.22±0.69	88.41±3.30	99.12±0.56	87.83±6.31	97.42±1.09	99.25±0.98
7	66.41±2.65	75.80±3.60	87.33±3.21	81.91±5.80	47.13±12.49	89.12±3.00	74.42±4.03	94.39±2.16	30.08±33.01	77.97±11.46	**99.99±0.03**
8	85.52±1.16	90.76±3.52	96.98±1.29	96.09±1.81	90.41±1.08	96.06±1.54	87.04±3.04	**97.64±1.14**	78.71±15.88	86.25±2.91	97.11±2.09
9	72.29±2.87	88.71±5.52	94.46±1.46	90.29±2.94	88.54±3.29	88.42±2.51	82.97±4.16	**96.55±0.90**	78.29±29.52	76.38±4.62	96.13±2.10
OA(%)	92.21±0.18	92.61±0.85	97.48±0.51	96.13±0.37	93.40±0.80	97.43±0.35	92.33±0.46	98.75±0.23	89.74±3.58	93.24±0.70	**99.24±0.42**
AA(%)	86.05±0.57	87.38±0.70	94.99±1.05	93.09±0.93	87.03±2.06	95.03±0.89	87.51±1.08	97.65±0.34	80.30±5.86	87.97±1.48	**98.82±0.50**
κ(%)	89.85±0.24	90.36±1.11	96.72±0.67	94.98±0.48	91.40±1.06	96.67±0.45	90.00±0.61	98.37±0.29	86.67±4.52	91.20±0.91	**99.01±0.55**
Params(k)	1484.6	387.8	538.4	133.3	483.2	4410.5	619.1	185.2	2263.2	71.4	555.3
FLOPs(M)	1260.7	3542.2	1315.5	440.9	2758.3	7516.0	2562.1	864.5	72.4	128.7	1927.9
Memory(MB)	24.8	78.2	61.4	56.9	151.0	135.0	162.4	53.1	71.7	148.5	38.8
Inference(ms)	0.37	1.97	3.81	2.48	4.12	3.31	4.29	1.73	0.98	8.41	7.45

TABLE VII

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE SA DATASET WITH 2% TRAINING SAMPLES

Class	2DCNN	3DCNN	SSFTT	morphFormer	spectralFormer	DCTN	GSC-ViT	MCMTN	MiniGCN	LG-GCN	Ours
1	99.73±0.23	99.35±0.56	**100±0**	99.66±1.00	99.84±0.28	**100±0**	94.38±8.75	99.59±1.34	51.24±44.61	99.83±0.17	**100±0**
2	97.26±0.53	98.97±0.68	99.99±0.03	99.87±0.16	98.85±0.54	**100±0**	99.12±0.98	100±0.01	90.28±22.12	99.47±0.26	**100±0**
3	94.88±0.48	98.30±0.55	99.92±0.20	97.07±2.72	95.64±1.86	99.83±0.08	95.28±1.05	**99.97±0.05**	46.45±41.68	99.39±0.37	99.95±0.13
4	96.37±0.29	97.06±0.83	99.44±0.43	96.18±1.46	96.64±0.85	96.55±2.81	97.20±1.01	99.19±0.65	92.06±24.16	95.75±0.84	**99.88±0.19**
5	99.31±0.45	98.46±0.73	99.82±0.27	97.90±1.34	97.15±1.27	99.72±0.27	98.78±1.25	99.54±0.48	**99.85±0.20**	98.94±0.65	99.32±0.38
6	**100±0**	100±0.01	99.98±0.05	99.96±0.12	99.98±0.07	**100±0**	99.98±0.03	**100±0**	100±0.01	99.98±0.03	**100±0**
7	98.85±0.09	99.09±0.18	99.97±0.03	99.79±0.23	99.75±0.15	100±0.01	99.46±0.33	99.98±0.01	98.20±3.47	99.59±0.20	**100±0**
8	91.50±1.29	88.05±2.13	95.33±1.37	91.06±3.12	88.18±1.90	96.05±1.53	88.67±2.75	95.83±1.76	92.38±5.38	95.89±1.19	**99.51±0.31**
9	99.15±0.16	99.19±0.40	99.95±0.10	99.91±0.19	99.82±0.18	**100±0**	98.05±0.85	100±0.01	99.98±0.02	99.98±0.05	**100±0**
10	97.14±0.47	97.65±0.65	98.39±1.03	97.07±1.24	96.37±1.43	97.80±0.56	93.02±4.56	97.94±1.47	97.74±3.63	97.49±1.26	**99.19±0.48**
11	89.54±1.46	90.99±1.48	98.69±1.45	97.89±2.36	91.80±2.92	**99.95±0.06**	91.51±6.19	99.39±1.20	37.82±40.58	98.12±1.57	99.19±0.81
12	99.95±0.07	**100±0**	99.99±0.03	99.94±0.12	99.92±0.16	**100±0**	99.94±0.08	100±0.02	46.91±49.35	99.99±0.04	**100±0**
13	98.72±0.34	98.96±0.51	99.88±0.27	98.41±0.75	98.33±0.93	99.65±0.19	95.50±4.53	**100±0**	53.46±43.25	99.26±0.47	99.61±0.53
14	**100±0**	98.13±1.16	99.90±0.18	99.66±0.88	**100±0**	**100±0**	99.87±0.19	99.99±0.03	69.54±37.25	**100±0**	99.77±0.26
15	93.94±1.24	79.07±6.50	96.20±2.50	92.74±5.39	92.47±5.88	95.40±1.68	82.51±8.74	96.86±2.56	49.35±34.96	96.59±1.13	**99.46±0.30**
16	96.03±1.01	94.57±1.07	99.72±0.38	97.09±3.60	96.95±3.24	**99.99±0.03**	82.65±6.73	99.27±1.53	77.14±35.41	99.93±0.16	99.76±0.29
OA(%)	96.20±0.23	93.79±0.91	98.35±0.45	96.47±1.18	95.51±0.98	98.32±0.31	93.40±1.67	98.49±0.79	81.90±5.26	98.22±0.38	**99.70±0.08**
AA(%)	97.02±0.17	96.12±0.53	99.20±0.22	97.76±0.88	96.98±0.60	99.06±0.23	94.75±2.00	99.22±0.52	75.15±9.39	98.76±0.22	**99.73±0.08**
κ(%)	95.77±0.25	93.07±1.01	98.16±0.50	96.07±1.32	95.00±1.10	98.13±0.35	92.63±1.87	98.32±0.88	79.61±6.00	98.02±0.43	**99.67±0.09**
Params(k)	1718.2	1210.0	1004.3	133.8	949.1	4412.3	619.5	215.2	4448.9	78.4	570.3
FLOPs(M)	2161.8	7014.0	2618.8	870.2	5646.7	7516.1	2562.1	1021.8	142.4	163.7	2004.9
Memory(MB)	28.7	143.5	100.3	108.8	168.2	147.8	179.9	58.8	99.2	177.4	41.0
Inference(ms)	0.33	3.86	2.13	1.97	4.91	3.11	4.28	1.82	1.13	8.84	7.32

between adjacent classes. On KSC, vegetation boundaries such as those between CP/oak and water exhibit fewer isolated misclassified pixels, showing improved continuity in gradual spectral transitions. On PU, material categories prone to confusion (e.g., gravel, bitumen, and shadows) present reduced fragmentation and more coherent spatial patterns compared with Transformer- and CNN-based baselines. On SA, agricultural parcels with irregular geometry—such as vineyard

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUALITATIVE RESULTS OF DIFFERENT METHODS ON THE HC DATASET WITH 0.6% TRAINING SAMPLES

Class	2DCNN	3DCNN	SSFTT	morphFormer	spectralFormer	DCTN	GSC-ViT	MCMTN	MiniGCN	LG-GCN	Ours
1	95.16±0.93	93.29±1.00	**97.97±0.25**	96.87±0.89	97.86±0.52	97.59±0.62	95.33±1.47	97.94±0.46	93.04±9.26	96.42±0.53	97.95±0.95
2	90.52±1.82	87.83±2.58	95.75±0.83	89.84±2.51	90.69±2.35	95.38±3.27	89.82±2.44	96.80±0.57	**97.08±3.51**	90.39±2.44	96.97±2.35
3	86.25±3.85	70.19±4.05	94.47±1.78	91.15±6.62	90.42±7.31	93.63±5.22	87.44±6.20	**98.25±1.44**	15.93±21.83	96.80±1.37	97.54±1.87
4	96.25±0.51	89.74±2.37	97.81±1.08	97.13±1.44	95.02±2.39	98.23±0.74	88.87±4.97	**98.53±0.56**	31.87±42.04	97.42±0.56	98.37±0.95
5	92.47±3.01	84.32±7.88	98.60±1.18	98.71±1.27	95.29±2.99	96.41±4.53	92.65±2.83	99.30±0.56	77.04±32.34	96.59±7.95	**99.83±0.34**
6	56.33±7.59	24.44±5.35	77.01±6.68	62.27±7.27	42.21±11.85	72.67±15.54	43.68±6.94	84.49±2.79	5.70±15.03	75.36±5.17	**88.03±2.77**
7	84.68±5.38	92.04±1.54	94.13±1.75	92.06±3.77	78.90±6.91	91.89±6.26	83.59±8.63	95.09±1.55	6.71±23.15	93.32±1.54	**96.93±1.32**
8	84.31±3.85	77.25±3.6	88.82±1.47	82.39±4.08	68.39±4.14	88.86±1.16	77.47±4.46	89.90±1.63	85.91±23.08	86.79±1.92	**91.76±2.01**
9	72.97±9.21	64.22±7.13	91.38±2.99	82.17±9.29	56.61±8.25	88.23±5.98	73.33±8.10	96.51±0.84	61.18±22.70	86.55±4.46	**97.29±1.14**
10	90.94±2.17	80.88±4.37	96.21±0.90	91.88±2.21	95.00±1.47	95.25±4.07	86.36±4.33	97.86±0.50	66.40±29.97	95.14±1.17	**98.40±0.67**
11	88.99±3.80	83.55±3.02	95.38±2.29	89.81±4.80	90.43±3.79	96.89±1.21	64.74±18.55	**98.46±0.65**	95.68±5.36	95.04±1.25	98.40±1.65
12	88.59±8.40	62.05±4.4	94.24±1.94	83.45±12.13	70.61±21.52	94.61±7.39	85.29±7.82	97.58±1.57	8.32±18.33	88.42±11.41	**98.95±1.19**
13	67.72±7.61	61.22±2.32	80.81±3.69	72.07±8.12	55.82±11.15	79.28±3.64	65.01±10.86	89.86±1.94	12.87±18.84	78.44±5.54	**92.29±1.93**
14	90.60±1.01	87.57±2.13	92.50±0.97	87.10±3.92	87.79±1.21	93.98±1.70	91.11±1.82	92.69±2.05	**97.96±2.54**	88.22±2.10	95.14±1.33
15	82.22±6.04	89.75±5.9	92.74±3.50	96.00±1.78	96.61±1.32	**98.82±0.87**	88.75±5.28	97.92±1.46	89.25±20.53	92.80±3.44	90.89±10.82
16	98.85±0.34	99.31±0.33	99.63±0.19	98.61±0.74	99.01±0.36	99.39±0.38	98.77±0.89	99.53±0.25	**99.85±0.30**	99.34±0.25	99.65±0.19
OA(%)	90.91±1.09	86.92±0.89	95.45±0.61	91.81±1.83	88.88±1.91	95.14±1.38	88.18±1.36	96.77±0.47	81.06±3.68	93.79±1.06	**97.33±0.60**
AA(%)	85.43±2.24	77.98±1.41	92.97±1.15	88.22±3.07	81.92±4.15	92.57±2.93	82.01±2.23	95.67±0.68	59.05±6.99	91.07±2.54	**96.15±1.15**
κ(%)	89.37±1.28	84.67±1.05	94.67±0.71	90.42±2.15	86.97±2.26	94.31±1.62	86.17±1.60	96.23±0.55	77.57±4.29	92.74±1.25	**96.88±0.70**
Params(k)	1879.4	1618.2	1326.8	133.8	1271.7	4412.3	619.5	235.4	5963.2	82.9	580.1
FLOPs(M)	2786.3	9431.7	3522.1	1167.6	7648.5	7516.1	2562.1	1130.8	190.8	187.9	2058.3
Memory(MB)	36.8	187.8	125.7	155.3	179.3	159.0	191.1	62.1	117.5	199.0	42.5
Inference(ms)	0.38	5.11	2.24	3.00	5.42	3.33	4.29	1.56	0.97	9.28	9.12

TABLE IX
ABLATION STUDY

#\Config	S³-Net		SDG	BVB	RENDER	KSC	PU	SA	HC
	DS Conv	Shiftwise							
BASE	✓	✓	✓	✓	✓	99.55	98.76	97.99	93.60
1A	×	×	✓	✓	✓	97.97 (↓1.58)	94.68 (↓4.08)	94.99 (↓3.00)	89.65 (↓3.95)
1B	×	✓	✓	✓	✓	99.43 (↓0.12)	96.81 (↓1.95)	95.46 (↓2.53)	89.29 (↓4.31)
1C	✓	×	✓	✓	✓	99.46 (↓0.09)	98.71 (↓0.05)	96.49 (↓1.50)	92.37 (↓1.23)
2A	✓	✓	×	×	×	98.73 (↓0.82)	94.13 (↓4.63)	91.97 (↓6.02)	88.03 (↓5.57)
2B	✓	✓	×	✓	✓	98.83 (↓0.72)	95.43 (↓3.33)	91.32 (↓6.67)	88.12 (↓5.48)
2C	✓	✓	✓	×	✓	98.52 (↓1.03)	94.64 (↓4.12)	93.02 (↓4.97)	87.91 (↓5.69)
2D	✓	✓	✓	✓	concat	98.81 (↓0.74)	93.96 (↓4.80)	93.87 (↓4.12)	87.99 (↓5.61)
3A	Transformer Blocks Replace CGABlocks					98.83 (↓0.72)	93.49 (↓5.27)	94.17 (↓3.82)	85.00 (↓8.60)
3B	Remove MHGConv					95.41 (↓4.14)	90.25 (↓8.51)	90.37 (↓7.62)	81.70 (↓11.90)

rows or fragmented crop plots—show more continuous shapes with mitigated local breaks. On HC, categories with limited training samples benefit from reduced scattered noise near-field boundaries, although slight artifacts remain in highly reflective regions. Overall, the qualitative trends align with the quantitative observations and indicate that the spatial arrangements produced by the model remain stable across different scene types.

D. Ablation Study

The ablation study evaluates the independent and joint contributions of the three core components in CentralGraphFormer: the spectral processing module (S^3-Net), the spatial processing module (SDG+BVB), and the CGAM. Based on Table IX, OA on four hyperspectral datasets (KSC, PU, SA, and HC) is reported relative to the full model (BASE). This study aims to clarify how each component supports the proposed "spectral purification–spatial screening–center-guided global refinement" pipeline described in Section III.

1) Ablation Analysis of the Spectral Processing Module: The spectral ablation focuses on the structural role of S^3-Net in modeling spectral redundancy and continuity. As shown in Table IX, completely removing the spectral processing module (1A) results in consistent and pronounced OA degradation across all datasets, with particularly large drops on PU and SA. This observation confirms that explicit spectral modeling is indispensable for discriminating spectrally similar classes.

To further examine the internal design of S^3-Net, configurations 1B and 1C selectively disable the depthwise spatial convolution and the shiftwise spectral operation, respectively. The results indicate that removing the depthwise convolution (1B) leads to moderate performance degradation, especially on spatially heterogeneous datasets, while removing the shiftwise operation (1C) causes more noticeable drops on SA and HC. This contrast suggests that the shiftwise mechanism plays a more critical role in capturing long-range spectral continuity, whereas depthwise convolution mainly contributes to local spatial–spectral coupling.

Overall, these results demonstrate that the two components of S^3-Net are complementary rather than redundant: depthwise convolution provides efficient local feature extraction, while shiftwise spectral shifting enables compact modeling of continuous interband correlations. Their joint design

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

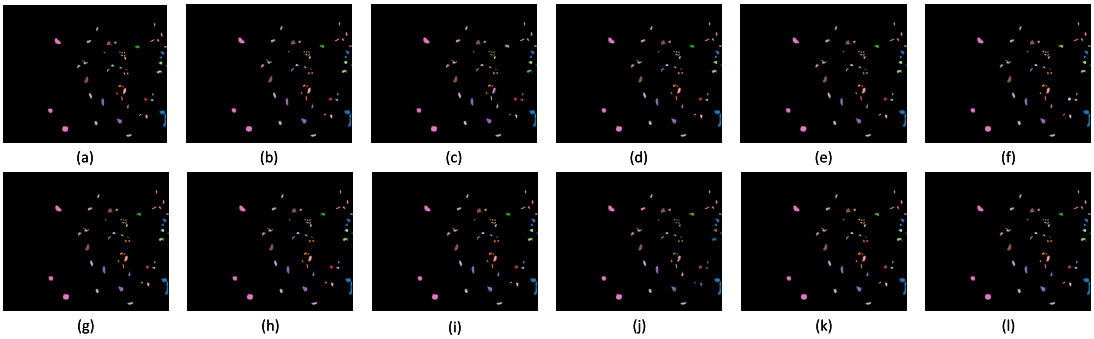


Fig. 6. Classification maps for the KSC dataset using 10% of training samples. (a) Ground truth. (b) 2DCNN. (c) 3DCNN. (d) SSFTT. (e) morphFormer. (f) spectralFormer. (g) DCTN. (h) GSC-ViT. (i) MCMTN. (j) MiniGCN. (k) LG-GCN. (l) Ours.

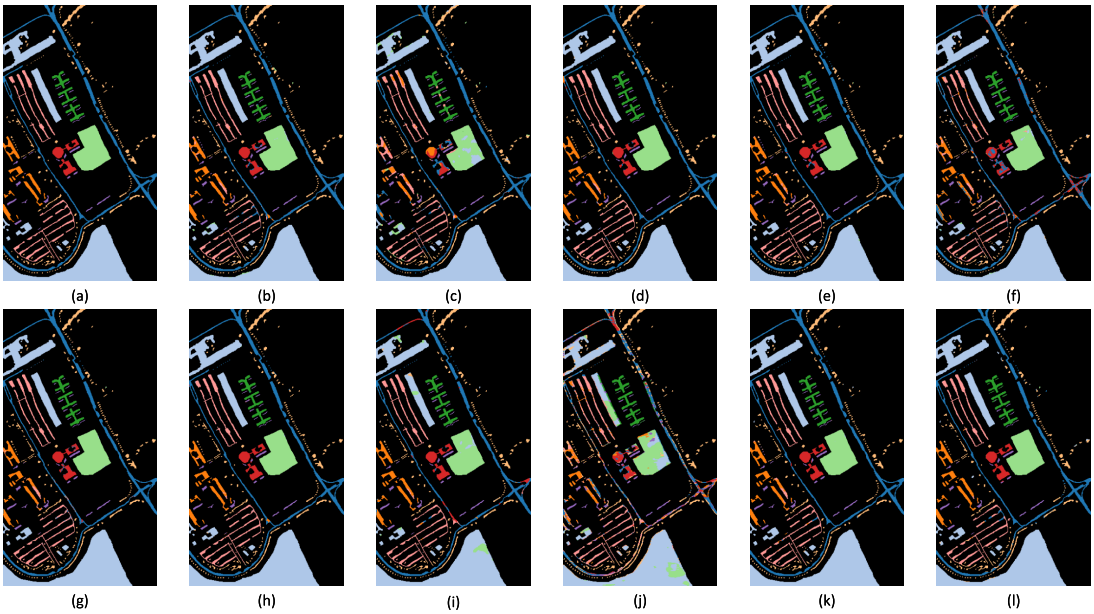


Fig. 7. Classification maps for the PU dataset using 2% of training samples. (a) Ground truth. (b) 2DCNN. (c) 3DCNN. (d) SSFTT. (e) morphFormer. (f) spectralFormer. (g) DCTN. (h) GSC-ViT. (i) MCMTN. (j) MiniGCN. (k) LG-GCN. (l) Ours.

allows effective spectral purification without introducing heavy attention or dense spectral mixing.

2) Ablation Analysis of the Spatial Processing Module: The spatial module ablation examines the roles of SDG and BVB in center-aligned spatial relevance estimation. Removing both SDG and BVB (2A) leads to substantial OA degradation across all datasets, particularly on PU and HC, indicating that naive patch-level processing is insufficient in heterogeneous scenes. When SDG is removed while retaining BVB (2B), performance drops remain significant, suggesting that stochastic relevance estimation is crucial for identifying reliable spatial neighborhoods under central-pixel supervision.

Conversely, removing BVB while retaining SDG (2C) also degrades performance, with pronounced drops on SA and HC, implying that smoothing and center-surround weighting contribute to stabilizing spatial relevance estimation. Replacing the proposed rendering strategy with naive feature concatenation (2-D) consistently underperforms the BASE model, confirming that explicit relevance-weighted rendering is more effective than static fusion.

TABLE X
CLASSIFICATION PERFORMANCE ON PU DATASET WITH DIFFERENT PATCH SIZES

Patch Sizes	SSFTT	DCTN	GSC-ViT	Ours
7×7	92.95±0.89	93.40±0.63	97.53±0.38	**97.72±0.72**
9×9	95.31±0.73	94.13±1.13	97.93±0.32	**98.46±0.43**
11×11	96.64±0.44	95.56±0.72	97.88±0.32	**98.84±0.48**
13×13	96.53±0.45	96.49±0.41	98.11±0.13	**98.92±0.24**
15×15	97.45±0.60	97.05±0.62	98.17±0.24	**98.56±0.79**
17×17	97.22±0.64	97.27±0.31	97.58±0.33	**98.64±0.78**
19×19	96.95±0.89	97.66±0.24	97.28±0.34	**99.20±0.40**

Collectively, these results show that SDG and BVB play complementary roles: SDG provides adaptive and center-aware spatial selection, while BVB regularizes relevance maps to improve robustness in complex or sample-scarce regions.

3) Ablation Analysis of the CGAM: To assess the contribution of center-guided graph reasoning, two CGAM variants are evaluated. In configuration 3A, the proposed CGAB is replaced by a standard Transformer block with dense self-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

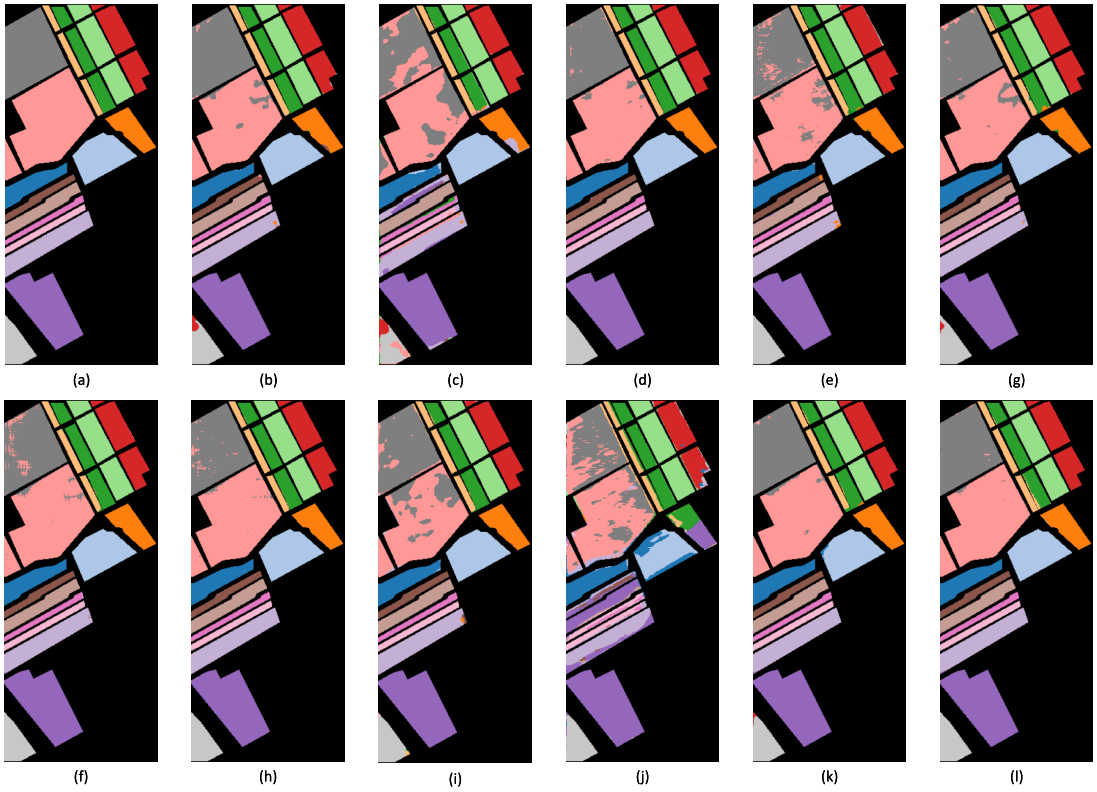


Fig. 8. Classification maps for the SA dataset using 2% of training samples. (a) Ground truth. (b) 2DCNN. (c) 3DCNN. (d) SSFTT. (e) morphFormer. (f) spectralFormer. (g) DCTN. (h) GSC-ViT. (i) MCMTN. (j) MiniGCN. (k) LG-GCN. (l) Ours.

TABLE XI
CLASSIFICATION PERFORMANCE ON HC DATASET WITH DIFFERENT PATCH SIZES

Patch Sizes	SSFTT	DCTN	GSC-ViT	Ours
7×7	83.11±0.95	87.85±0.43	90.78±0.34	**91.80±0.49**
9×9	87.67±0.76	89.82±0.37	93.14±0.63	**93.68±0.42**
11×11	90.23±0.83	91.41±0.15	93.85±0.55	**94.77±0.42**
13×13	92.13±0.83	93.07±0.65	94.55±0.45	**95.58±0.48**
15×15	93.18±0.72	93.78±0.27	95.37±0.57	**96.26±0.25**
17×17	94.31±0.51	94.61±0.36	95.86±0.36	**96.75±0.48**
19×19	94.67±0.71	95.81±0.19	95.91±0.28	**96.90±0.36**

attention. This replacement leads to consistent performance degradation across all datasets, particularly on PU and HC, indicating that generic self-attention is less suitable for pixel-level hyperspectral classification, where center relevance must be explicitly modeled.

Configuration 3B removes the MHGConv from CGAB, leaving only tokenwise feed-forward refinement. This modification causes the most severe performance drops among all ablation settings, confirming that explicit relational modeling between the central pixel, and its informative neighbors is essential. Without graph-based message passing, the model fails to propagate discriminative contextual cues effectively.

These observations demonstrate that CGAM's advantage does not stem merely from Transformer-style refinement, but from its center-anchored sparse graph formulation, which enables efficient and targeted global reasoning aligned with the classification objective.

4) Comprehensive Discussion and Research Implications: The ablation results reveal a clear hierarchy of functional contributions within CentralGraphFormer. The CGAM serves as the primary performance driver by enabling center-guided relational reasoning, while the spectral processing module provides essential spectral purification by compactly modeling continuous band dependencies. The spatial processing module further enhances robustness by adaptively screening and stabilizing spatial context under central-pixel supervision.

Importantly, the results validate the proposed design philosophy that global reasoning should be explicitly centered on the target pixel rather than uniformly applied over patches. Dense self-attention or naive feature fusion fails to provide this inductive bias, whereas the proposed combination of spectral continuity modeling, biologically inspired spatial relevance estimation, and center-driven graph refinement yields a more interpretable and efficient solution.

These findings suggest broader implications for hyperspectral image analysis: effective pixel-level classification benefits from prioritizing center-aware global modeling while maintaining lightweight and complementary spectral–spatial processing. Integrating structured graph reasoning with principled inductive biases offers a promising direction for balancing performance, efficiency, and interpretability in future HSI models.

E. Parameters Analysis

This section evaluates the sensitivity of the proposed framework to three key hyperparameters: the number of graph

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.



Fig. 9. Classification maps for the HC dataset using 0.6% of training samples. (a) Ground truth. (b) 2DCNN. (c) 3DCNN. (d) SSFTT. (e) morphFormer. (f) spectralFormer. (g) DCTN. (h) GSC-ViT. (i) MCMTN. (j) MiniGCN. (k) LG-GCN. (l) Ours.

convolutional heads, the spatial patch size, and the training sample ratio. These experiments examine how the model behaves under structural changes (heads), contextual variations (patch size), and data scarcity (training samples), providing practical guidance for parameter selection while assessing robustness.

1) Analysis of the Number of Heads: We first investigate the effect of the number of graph convolution heads (1-32) using a fixed patch size of 13×13 and 32/16 training/validation samples per class. As shown in Fig. 10, the optimal number of heads varies with spectral dimensionality: datasets with more spectral bands generally benefit from more heads. KSC and PU reach their best performance with 2-4 heads, whereas the higher dimensional SA and HC datasets achieve optimal accuracy with 8–16 heads.

Training time grows nearly linearly with the number of heads. Considering both accuracy and computational cost, eight heads consistently provide near-optimal performance across datasets while keeping the training time moderate. Thus, eight heads are adopted as the default configuration in subsequent experiments.

2) Analysis of Patch Sizes: Patch size determines the spatial contextual range. We compare sizes from 7×7 to 19×19 on top-performing models (SSFTT, DCTN,GSC-ViT, and Ours). The impact of different parameter settings on classification performance is illustrated in Fig. 11. Results in Tables X and XI show a clear trend: larger patches lead to higher accuracy due to richer spatial cues. Performance improvements become less pronounced beyond 13×13, suggesting diminishing returns with larger context.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

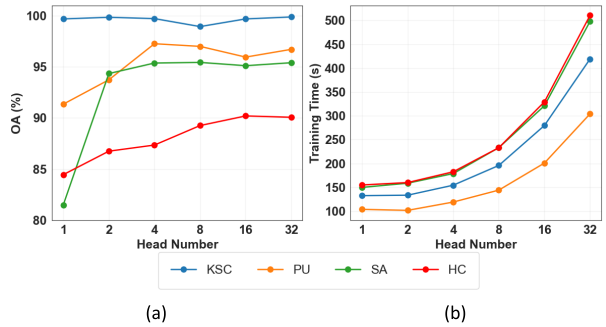


Fig. 10. OA and training time with different numbers of graph convolution heads. (a) OA. (b) Training time.

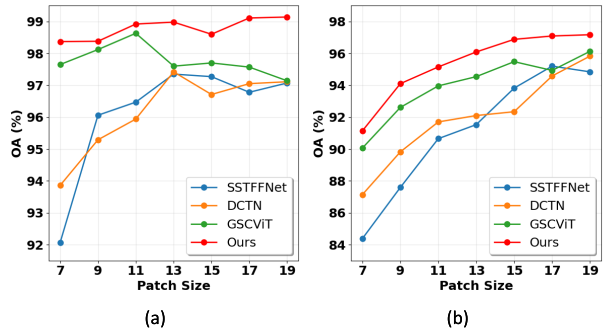


Fig. 11. OA with different patch sizes. (a) PU dataset. (b) HC dataset.

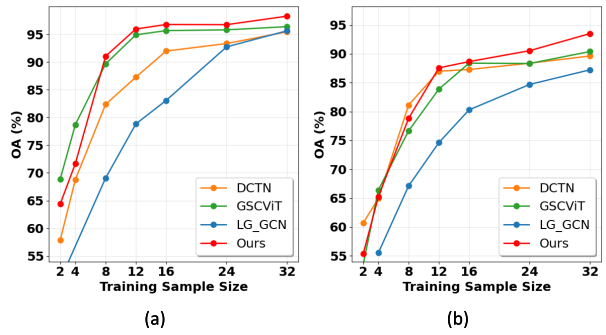


Fig. 12. OA with different training sample sizes. (a) PU dataset. (b) HC dataset.

Across all settings, the proposed model consistently ranks first. Its advantage becomes more evident under larger patches, indicating its ability to effectively utilize expanded spatial information. Overall, 13×13 offers a balanced choice between performance and cost, while 19×19 provides the best accuracy when computation is not a limiting factor.

3) Analysis of Training Sample Sizes: Using the default configuration (eight heads, 13 × 13 patches), we assess model robustness under varying training samples (2–32per class). The impact of different parameter settings on classification performance is illustrated in Fig. 12. As shown in Tables XII and XIII, the model maintains competitive accuracy even in extremely low-sample regimes. Its performance rises rapidly with increasing samples and surpasses other methods once the sample size reaches moderate levels (8–12 per class).

TABLE XII

CLASSIFICATION PERFORMANCE ON PU DATASET WITH DIFFERENT TRAINING SAMPLE SIZES

Samples per class		Methods			
Train	Val	DCTN	GSC-ViT	LG-GCN	Ours
2	1	57.88±8.97	**68.88±2.64**	50.65±11.90	64.40±4.57
4	2	68.79±4.77	**78.71±6.33**	47.45±13.23	71.68±2.65
8	4	82.36±3.03	89.66±4.91	69.06±2.44	**91.04±3.10**
12	6	87.26±1.59	94.90±1.54	78.83±3.93	**95.95±0.95**
16	8	91.97±0.80	95.66±0.69	83.06±2.72	**96.76±1.37**
24	12	93.33±1.80	95.80±1.03	92.73±1.38	**96.73±1.92**
32	16	95.45±0.64	96.37±0.51	95.63±0.78	**98.27±0.58**

TABLE XIII

CLASSIFICATION PERFORMANCE ON HC DATASET WITH DIFFERENT TRAINING SAMPLE SIZES

Samples per class		Methods			
Train	Val	DCTN	GSC-ViT	LG-GCN	Ours
2	1	**60.71±1.71**	53.49±8.11	45.28±6.15	55.35±2.22
4	2	64.92±1.90	**66.39±5.06**	55.51±3.58	65.23±5.63
8	4	**81.13±1.21**	76.65±3.52	67.10±1.91	78.82±0.91
12	6	86.98±0.60	83.88±4.19	74.61±2.23	**87.56±1.21**
16	8	87.28±1.62	88.38±0.63	80.29±1.43	**88.65±0.65**
24	12	88.37±1.25	88.32±1.58	84.69±1.87	**90.54±1.54**
32	16	89.62±1.09	90.40±2.50	87.23±0.57	**93.51±1.13**

On both PU and HC, the proposed model shows the steepest improvement curve and achieves the highest accuracy at 32 samples per class, without signs of saturation. This indicates strong data efficiency and adaptability to both few-shot and fully supervised conditions. These characteristics make the model particularly suitable for real-world tasks, where labeled hyperspectral samples are scarce.

V. CONCLUSION

This article presented CentralGraphFormer, a unified hyperspectral image classification framework inspired by biological visual perception and designed to address key limitations in existing CNN-, Transformer-, and hybrid-based models. While convolutional networks excel at local spatial–spectral representation and Transformer architectures enable long-range interaction modeling, many existing approaches—especially hybrid designs—still rely on uniform patch-level processing and shallow fusion strategies. Such formulations overlook the center-pixel decision principle inherent to pixelwise HSI classification and limit the effective integration of local and global cues.

CentralGraphFormer addresses these issues through a coordinated, center-guided processing pipeline. The S^3-Net efficiently captures spectral continuity and long-range band correlations, providing compact and discriminative spectral representations. The SDG and BVB modules introduce adaptive, center-aligned spatial relevance screening, selectively emphasizing informative contextual pixels while suppressing distractors. Building upon these refined features, the CGAM performs sparse, center-focused relational reasoning, enabling the interpretable global interaction modeling with substantially reduced computational overhead compared with dense self-attention mechanisms. Together, these components establish

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

a hierarchical feature processing strategy that progressively refines spectral information, adapts spatial context, and conducts targeted global reasoning.

Extensive experiments on multiple benchmark datasets demonstrate that the proposed framework achieves competitive accuracy, strong stability, and favorable parameter efficiency. Ablation studies further reveal a clear hierarchy of component contributions, highlighting the central role of center-guided graph reasoning and validating the necessity of explicitly aligning global modeling with the classification objective. Beyond the specific architecture, this work suggests a broad.

REFERENCES

[1] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[2] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509711.

[3] J. Pontius, M. Martin, L. Plourde, and R. Hallett, "Ash decline assessment in emerald ash borer-infested regions: A test of tree-level, hyperspectral technologies," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 2665–2676, May 2008.

[4] Z. Zhang et al., "UAV hyperspectral remote sensing image classification: A systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3099–3124, 2025.

[5] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[6] X. Sun, B. Wang, Z. Wang, H. Li, H. Li, and K. Fu, "Research progress on few-shot learning for remote sensing image interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2387–2402, 2021.

[7] B. Lu, P. Dao, J. Liu, Y. He, and J. Shang, "Recent advances of hyperspectral imaging technology and applications in agriculture," *Remote Sens.*, vol. 12, no. 16, p. 2659, Aug. 2020.

[8] M. Sheykhmousa, M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni, "Support vector machine versus random forest for remote sensing image classification: A meta-analysis and systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6308–6325, 2020.

[9] A. Villa, J. A. Benediktsson, J. Chanussot, and C. Jutten, "Hyperspectral image classification with independent component discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4865–4876, Dec. 2011.

[10] W. Liao, A. Pizurica, P. Scheunders, W. Philips, and Y. Pi, "Semisupervised local discriminant analysis for feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 184–198, Jan. 2013.

[11] Q. Wang, Z. Meng, and X. Li, "Locality adaptive discriminant analysis for spectral–spatial classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2077–2081, Nov. 2017.

[12] B. Schölkopf and A. J. Smola, *Learning With Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. Cambridge, MA, USA: MIT Press, 2002.

[13] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[14] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based k -nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov. 2010.

[15] C. Lou et al., "Land use/land cover (LULC) classification using hyperspectral images: A review," *Geo-spatial Inf. Sci.*, vol. 28, no. 2, pp. 345–386, 2024.

[16] H. Lee and H. Kwon, "Contextual deep CNN based hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Beijing, China, Jul. 2016, pp. 3322–3325.

[17] Y. Garini, I. T. Young, and G. McNamara, "Spectral imaging: Principles and applications," *Cytometry A*, vol. 69A, no. 8, pp. 735–747, Aug. 2006.

[18] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[19] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, 2018.

[20] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, and A. Abbas, "Modern trends in hyperspectral image analysis: A review," *IEEE Access*, vol. 6, pp. 14118–14129, 2018.

[21] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[22] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[23] L. Gao, S. Xiao, C. Hu, and Y. Yan, "Hyperspectral image classification based on fusion of convolutional neural network and graph network," *Appl. Sci.*, vol. 13, no. 12, p. 7143, Jun. 2023.

[24] C. Shi, H. Wu, and L. Wang, "CEGAT: A CNN and enhanced-GAT based on key sample selection strategy for hyperspectral image classification," *Neural Netw.*, vol. 168, pp. 105–122, Nov. 2023.

[25] Y. Gan, S. Wu, X. Li, Z. Xu, and Y. Pan, "Graph-aware hybrid encoding for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, 2025, Art. no. 5509105.

[26] J. Zhang, B. Tu, B. Liu, J. Li, and A. Plaza, "Hyperspectral image classification via neighborhood adaptive graph isomorphism network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515717.

[27] Q. Hong et al., "SATNet: A spatial attention based network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 22, p. 5902, Nov. 2022.

[28] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715.

[29] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[30] X. He, Y. Chen, and Q. Li, "Two-branch pure transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5902.

[32] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.

[33] J. An, L. Dai, M. Wang, W. Zhang, and X. Zhang, "SPGFormer: Structure perception graph transformer with Laplacian position encoding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5521517.

[34] H. Wu et al., "CvT: Introducing convolutions to vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 22–31.

[35] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.

[36] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[37] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[38] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[39] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.

[40] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.

[41] X. Yang, W. Cao, D. Tang, Y. Zhou, and Y. Lu, "Actn: Adaptive coupling transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, 2025, Art. no. 5503115.
[42] F. Zhao, J. Zhang, Z. Meng, H. Liu, Z. Chang, and J. Fan, "Multiple vision architectures-based hybrid network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 234, Dec. 2023, Art. no. 121032.
[43] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.
[44] B. Xi et al., "MCTGCL: Mixed CNN–transformer for Mars hyperspectral image classification with graph contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503214.
[45] T. Wang, Y. Sun, and Y. Hu, "Multiscale graph transformer network with dynamic superpixel pyramid for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13438–13456, 2025.
[46] W. Chen et al., "REST: Holistic learning for end-to-end semantic segmentation of whole-scene remote sensing imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 1, pp. 693–710, Jan. 2026.
[47] Z. Meng, J. Zhang, F. Zhao, H. Liu, and Z. Chang, "Residual dense asymmetric convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 3159–3162.
[48] J. Zhou, S. Zeng, G. Gao, Y. Chen, and Y. Tang, "A novel spatial–spectral pyramid network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.
[49] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.
[50] X. Liang, W. Liu, Y. Zhang, J. Yu, and H. Qu, "Dimensionality-varied convolutional neural network for hyperspectral image classification with small-sized labeled samples," in *Proc. 9th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Sep. 2018, pp. 1–5.
[51] B. Cui, X.-M. Dong, Q. Zhan, J. Peng, and W. Sun, "LiteDepthwiseNet: A lightweight network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.
[52] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.
[53] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.
[54] J. Chen, W. Li, and H. El-Askary, "Spectral–spatial transformer with multi-scale convolutional attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 24223–24241, 2025.
[55] Y. Wang, M. Tai, Z. Liu, Z. Shu, F. Xiong, and Z. Yu, "Mid-range convolutional modulated transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.
[56] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
[57] Y. Wang, X. Yu, H. Dong, and S. Zang, "Graph convolutional network with local and global feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536915.



Zilin Zhang received the B.S. degree in automation from Harbin Engineering University, Harbin, China, in 2024. He is currently pursuing the Ph.D. degree in control science and engineering with Central South University, Changsha, China.

His research interests mainly focus on computational spectral imaging, image processing, and computer vision.



Haoyang Yu received the B.S. and Ph.D. degrees in precision instrument from Tsinghua University, Beijing, China, in 2016 and 2022, respectively.

Since February 2022, he has been with the Department of Automation, Central South University, Changsha, China, where he is currently a Lecturer. His research interests include optical measurement, laser spectroscopy, laser ranging, and signal processing.



Zhaohui Jiang (Member, IEEE) received the M.Eng. degree in automatic control engineering and the Ph.D. degree in control science and engineering from Central South University, Changsha, China, in 2006 and 2011, respectively.

He is currently a Professor with Central South University. His research interests include detection technology and automatic equipment, image processing, industrial VR, and modeling and optimal control of complex industrial processes.



Kangkang Li received the B.Eng. degree from Zhejiang University of Technology, Hangzhou, China, and the M.Eng. degree from Central South University, Changsha, China, in 2022 and 2025, respectively.

His main research interests are hyperspectral image processing and machine learning.



Dong Pan (Member, IEEE) received the B.S. degree in automatic and the Ph.D. degree in control science and engineering from Central South University, Changsha, China, in 2015 and 2021, respectively.

He is currently an Associate Professor with Central South University. His main research interests include infrared thermography, vision-based measurement, image processing, and deep learning.



Weihua Gui (Member, IEEE) received the degree of the B.Eng. degree in electrical engineering and the M.Eng. degree in automatic control engineering from Central South University, Changsha, China, in 1976 and 1981, respectively.

From 1986 to 1988, he was a Visiting Scholar with the University of Duisburg-Essen, Duisburg, Germany. He has been a Full Professor with Central South University, since 1991. His main research interests are in modeling and optimal control of complex industrial process, distributed robust control, and fault diagnoses.

Mr. Gui is an Academician of Chinese Academy of Engineering.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:43 UTC from IEEE Xplore. Restrictions apply.